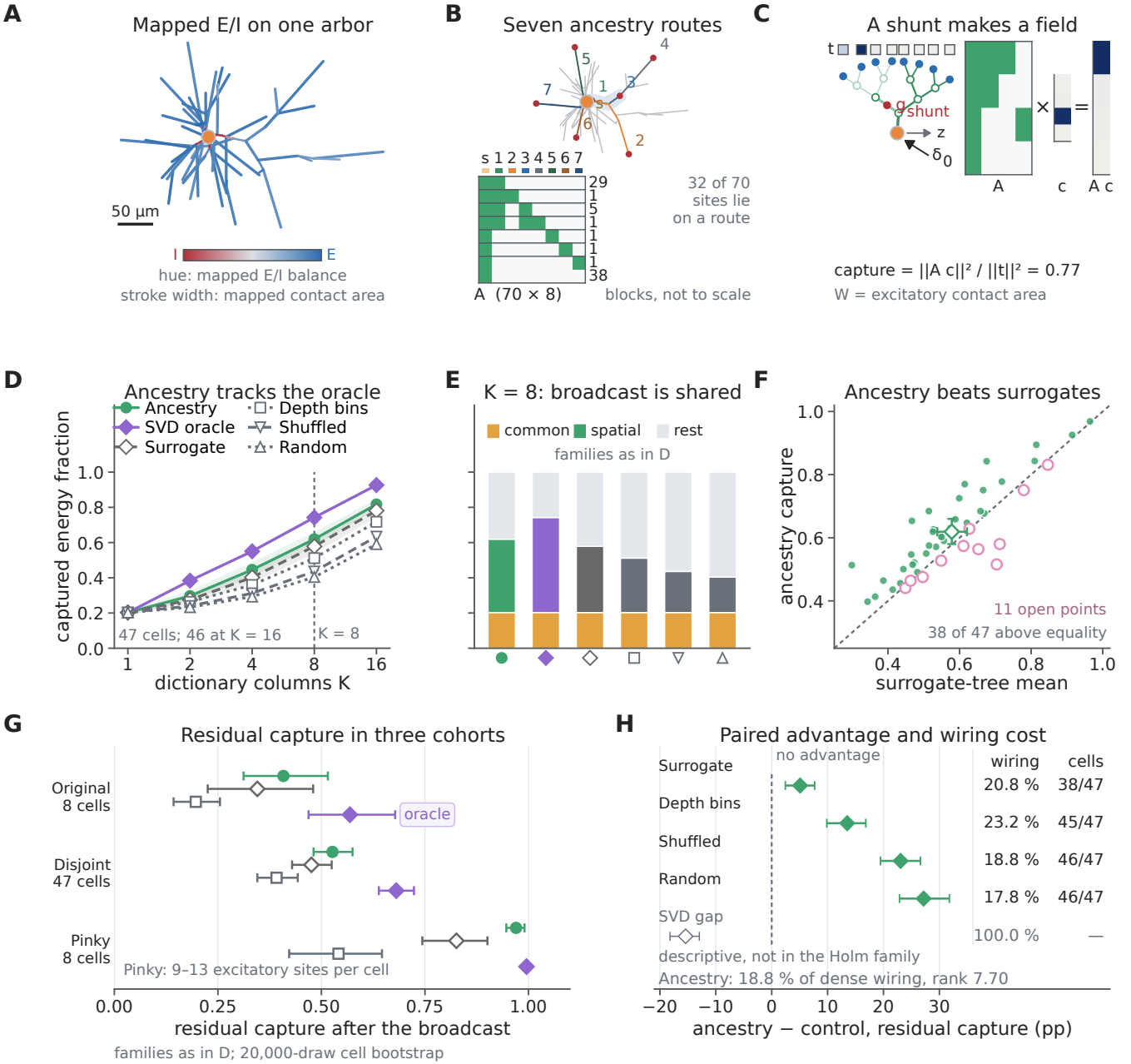

A
Mapped E/I on one arbor
50 µm
I
E
hue: mapped E/I balance
stroke width: mapped contact area
B
Seven ancestry routes
s 1 2 3 4 5 6 7
29
1
5
1
1
1
1
38
A (70 × 8)
32 of 70
sites lie
on a route
blocks, not to scale
C
A shunt makes a field
t
g shunt
z
δ0
A
c
A c
capture = ||A c||² / ||t||² = 0.77
W = excitatory contact area
D
Ancestry tracks the oracle
Ancestry
SVD oracle
Surrogate
Depth bins
Shuffled
Random
captured energy fraction
47 cells; 46 at K = 16
K = 8
dictionary columns K
E
K = 8: broadcast is shared
common
spatial
rest
families as in D
F
Ancestry beats surrogates
ancestry capture
surrogate-tree mean
11 open points
38 of 47 above equality
G
Residual capture in three cohorts
Original
8 cells
oracle
Disjoint
47 cells
Pinky
8 cells
Pinky: 9–13 excitatory sites per cell
residual capture after the broadcast
families as in D; 20,000-draw cell bootstrap
H
Paired advantage and wiring cost
no advantage
wiring
cells
Surrogate
20.8 %
38/47
Depth bins
23.2 %
45/47
Shuffled
18.8 %
46/47
Random
17.8 %
46/47
SVD gap
100.0 %
—
descriptive, not in the Holm family
Ancestry: 18.8 % of dense wiring, rank 7.70
ancestry − control, residual capture (pp)